AI & Machine Learning — Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning

1. Objective

The objective is to assess model reliability rather than relying only on training accuracy. This work detects overfitting using training and test RMSE, estimates generalization using 5-fold cross-validation, tunes Decision Tree hyperparameters with GridSearchCV, and justifies the final model.

2. Dataset and preprocessing

The provided California Housing CSV contains 20,640 rows and 10 columns. The target is **median_house_value**. All other columns are used as input features, including numeric location/demographic/housing variables and the categorical **ocean_proximity** feature. The data is split into 80% training and 20% test data with random_state=42. Numeric missing values are imputed with the median and scaled using StandardScaler; categorical missing values are filled with the most frequent category and encoded using OneHotEncoder. Preprocessing is placed inside a pipeline to prevent information leakage from the held-out test set.

3. Validation and tuning approach

Linear Regression, Ridge Regression (alpha=1.0), and a baseline Decision Tree are compared. Each model is evaluated using test RMSE, test R^2, and 5-fold cross-validation RMSE on the training set. GridSearchCV tunes a Decision Tree across max_depth = [3, 5, 7, 10, None], min_samples_split = [2, 5, 10], and min_samples_leaf = [1, 2, 5]. The selection metric is negative root mean squared error, so the best search result corresponds to the lowest RMSE.

4. Model comparison results

Model	Train RMSE	Test RMSE	Test R^2	5-fold CV RMSE
Linear Regression	68,434	70,059	0.625	68,623
Ridge Regression	68,435	70,066	0.625	68,622
Decision Tree (baseline)	0	69,176	0.635	69,791
Tuned Decision Tree	49,956	60,181	0.724	61,039

RMSE is expressed in the dataset's house-value units; lower is better. R^2 measures the proportion of target variance explained on the held-out test set; higher is better. Cross-validation results are computed from the training partition only.

5. Overfitting analysis

The baseline Decision Tree obtains a training RMSE of 0, which is effectively zero, but its test RMSE is 69,176. This large train-test gap is evidence of overfitting: the unrestricted tree memorizes the training examples but does not generalize as well to unseen records.

After tuning, the tree's training RMSE rises to 49,956, while test RMSE improves to 60,181; test R^2 improves to 0.724. This is the expected trade-off when model complexity is constrained: training fit becomes less perfect, but performance on unseen data improves.

6. Best hyperparameters

GridSearchCV selected: max_depth = 10, min_samples_leaf = 5, min_samples_split = 2. The best 5-fold cross-validation RMSE is 61,039.

7. Final model selection justification

The Tuned Decision Tree is selected because it provides the lowest test RMSE and the highest test R^2 among the four evaluated models, while its cross-validation RMSE is substantially lower than the baseline tree's. Cross-validation is preferred over a single split for model selection because it averages results across multiple folds. The selected depth and leaf-size constraints reduce overfitting while retaining non-linear relationships in housing data. Linear and Ridge Regression are simpler and provide useful baselines, but their test errors are higher in this experiment.

8. Limitations and reproducibility

Results are specific to the provided CSV, random seed, preprocessing, and search grid. The test set should remain untouched during hyperparameter selection. The California Housing target has a known upper cap, so errors for the highest-value homes may be affected by that dataset property. To reproduce the results, run all cells in *AI_ML_Task3_Model_Validation_Tuning.ipynb* with the CSV in the same folder and Python packages pandas, numpy, matplotlib, and scikit-learn installed.

Deliverables included

1. Jupyter Notebook with preprocessing, baseline evaluation, cross-validation, GridSearchCV, model comparison, and plots.
2. Model comparison CSV.
3. Hyperparameter search results CSV.
4. This explanatory report.